

BIDIRECTIONAL BLE ARCHITECTURE • RTLS

Track More, Install Less

Room-level visibility across complex indoor environments, without a gateway in every room. BiLink relay anchors extend BLE coverage into private rooms, closets and transitional spaces, and report through your existing ZENIX backbone into SOLIX.



THE BILINK ANCHOR

NODIX CEN-1

BiLink BLE Relay Anchor
Battery-operated scanner and beacon



2–4 m

Room-level
accuracy *



Up to 70%

Lower infrastructure
cost per room *



2–3 y

Battery life,
replaceable cells *



Zero

Cabling and IP
addresses per room

One scanner per floor cannot see **every room**



THE PROBLEM

In large facilities with rooms of varying size and layout, a single scanner cannot cover the whole area. Blind spots appear, signal blends through walls, and asset movement goes undetected, costing response time and operational confidence. Placing a cabled gateway in every room removes the blind spots but multiplies cabling, IP addresses, maintenance and cost.



THE BILINK ANSWER

Bilink splits the job. High-bandwidth ZENIX gateways stay in hallways and open walls and carry the data backbone. Battery-powered NODIX CEN-1 anchors sit inside the rooms, scan locally, and relay what they see back to the nearest gateway over BLE. Presence becomes a fact, not an inference: if the anchor in Room 101 sees a tag, the tag is in Room 101.



Hybrid deployment model

GATEWAYS IN THE HALLS • ANCHORS IN THE ROOMS

ROOM 101



NODIX CEN-1 scans locally



BLE relay

ROOM 102



NODIX CEN-1 scans locally



BLE relay

EXAM BAY



NODIX CEN-1 scans locally



BLE relay

STORE ROOM



NODIX CEN-1 scans locally



BLE relay



HALLWAY • ZENIX LEN gateway • PoE / Wi-Fi backbone

One gateway serves many anchors



ETHERNET / WI-FI



SOLIX positioning engine

HOW IT WORKS



01

Local scan

The anchor scans its room for BLE tags on equipment, patients and staff.



02

Wireless bypass

It bundles tag data with its own Room ID and relays it to the nearest ZENIX gateway.



03

Bidirectional validation

Every anchor both scans and broadcasts, validating presence and smoothing zone handover.



04

Real-time intelligence

SOLIX resolves the room, publishes position and fire's zone entry and exit events.

The same room-level certainty, a fraction of the install

A worked example: a 40-room ward where every room needs confirmed presence, not hallway inference.

40-ROOM WARD	GATEWAY IN EVERY ROOM	BILINK HYBRID
 Devices deployed	40 cabled gateways	4 ZENIX gateways + 40 NODIX CEN-1 anchors
 Cabling runs	40 PoE drops, one per room	4 — anchors mount with tape or a screw
 Network footprint	40 IP addresses to provision and secure	4 IP addresses; anchors add no network overhead
 Installation labour	Electrician time per room, celling access, downtime	Minutes per room, no trades, no ward closure
 Hardware cost per room	Gateway + cabling + labour	Up to 70% lower*
 Total deployment cost	Hardware plus cabling, labour and ward downtime	Saving is larger still — the per-room cabling and labour disappear
 Room-level certainty	Yes, with signal bleed between walls	Yes, presence confirmed in the room itself
 Adding a room later	New cable run and switch port	Mount one anchor, provision it in SDM

* Hardware only. Because BLink also removes the per-room cabling, networking and installation labour, the saving on total deployment cost is higher again. Indicative figures based on typical European rates; actual figures depend on site conditions and local labour cost.

KEY FEATURES & BENEFITS



Reliable room-level visibility

Consistent presence detection in rooms and defined zones using calibrated BLE thresholds.



Extended coverage

Relay anchors reach small rooms, entry points and transitional spaces without interruption.



Reduced infrastructure

Lower gateway dependency cuts cabling, network overhead and total deployment cost.



Bidirectional BLE operation

Anchors scan and broadcast at once, so detection is continuously exchanged and validated.



Battery-powered deployment

Replaceable high-capacity cells and zero-cable mounting allow fast, scalable rollout.



Entry & exit detection

Movement between zones is tracked through continuous handoff and stable detection.



Healthcare / Hospitals

Room-level tracking of patients, staff and equipment without hallway bleed. Better handovers and reliable entry/exit monitoring in sensitive areas.



Warehouse / Logistics

Zone tracking of assets and personnel with fewer gateways. Improved visibility, optimised infrastructure, entry and exit alerting.



Smart Buildings

Room-level occupancy and movement insight for space utilisation, secure access monitoring and lower infrastructure cost.

WHAT A BILINK DEPLOYMENT INCLUDES



NODIX CEN-1
BiLink BLE relay anchor, battery-operated scanner and beacon



ZENIX LEN-1
BLERSSI indoor gateway with PoE and Wi-Fi



PINIX Tags
Asset and person tags with SOS button and temperature sensor



Solix RTLS & IoT
Positioning engine, device management and visualisation, cloud on-premise

For system integrators

- REST & WebSocket APIs** — poll or subscribe to live position, zone entry/exit and sensor events from the SOLIX engine.
- MQTT streaming** — publish device and location telemetry into your own broker and event pipeline.
- omlox compatible** — interoperate with other location providers in a shared deployment.
- ERP, BMS & clinical systems** — middleware and iFrame embedding for SAP, building management and hospital platforms.
- Cloud or on-premise** — the same platform and API surface either way; no vendor lock on hosting.
- Field provisioning** — configure anchors and tags from the SDM app on Android and iOS, PIN protected.




 Also iBeacon, Eddystone and sBeacon advertising

NODIX CEN-1 at a glance

	Radio	Bluetooth Low Energy 5.x, 2.4 GHz
	Advertising	iBeacon, Eddystone (UID & TLM), sBeacon
	Adv. / scan interval	100 ms – 10.2 s / 1000 ms – 5 s
	No. Scanned Tags	upto 12 BLE Tags (w/ highest RSSI signal)
	Transmit power	+4 dBm to –20 dBm
	Antenna	PCB, omnidirectional
	Sensors	Internal temperature; accelerometer (optional); button, LED
	Battery	2 × ER14505, 4800 mAh, replaceable
	Battery life	1 y 1 m at 1000 ms; 2 y 2 m at 5000 ms*
	Enclosure	White ABS, 70.95 mm Ø × 26.20 mm, 72 g, IP54; tape or screw mount
	Environment	–20 °C to 70 °C, up to 90% humidity
	Certification	CE; FCC pending

* Estimated with sBeacon and iBeacon enabled only.

Eliminate blind spots. Track with confidence.

Book a technical session with our engineers, or request an evaluation kit to validate BiLink on your own floor plan.

✉ sales@sentrax.com

☎ +41 41 505 0065

🌐 www.sentrax.com

📍 Sentrax GmbH · Root D4, Luzern, Switzerland



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Smart Sensing &
Locating Systems
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